

CO1-Output Prediction

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CSA0307 Data Structure

SIMATS Online Programming Lab

CO1-Output Prediction

QUESTIONS

- Q1 Largest and Smallest element 1 mark
- Q2 Second Largest Element 1 mark
- Q3 Occurrence of an element 1 mark

3/3 answered

Q1 Largest and Smallest element 1 mark

Write a C program for Find the Largest and Smallest Element in an Array

Your Code

```
#include <stdio.h>
int main() {
    int n, i, a[100];
    scanf("%d", &n);
    for (i = 0; i < n; i++) scanf("%d", &a[i]);
    int min = a[0], max = a[0];
    for (i = 1; i < n; i++) {
        if (a[i] < min) min = a[i];
        if (a[i] > max) max = a[i];
    }
    printf("Largest: %d\nSmallest: %d\n", max, min);
    return 0;
}
```

Input

5
12 5 34 89 102

Output

Largest: 102
Smallest: 5

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QUESTIONS

- Q1 Largest and Smallest element 1 mark
- Q2 Second Largest Element 1 mark
- Q3 Occurrence of an element 1 mark

3/3 answered

Q2 Second Largest Element 1 mark

Write a C program for find the second largest element

Your Code

```
#include <stdio.h>
int main() {
    int n, i, a[100];
    scanf("%d", &n);
    for (i = 0; i < n; i++) scanf("%d", &a[i]);
    int max1 = a[0], max2 = -1;
    for (i = 1; i < n; i++) {
        if (a[i] > max1) {
            max2 = max1;
            max1 = a[i];
        } else if (a[i] > max2) {
            max2 = a[i];
        }
    }
    printf("Second Largest: %d\n", max2);
    return 0;
}
```

Input

5
12 35 1 89 25

Output

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- Q1 Largest and Smallest element 1 mark
- Q2 Second Largest Element 1 mark
- Q3 Occurrence of an element 1 mark

3/3 answered

Q3 Occurrence of an element 1 mark

Write a C program for Find the First and Last Occurrence of an Element

Your Code

```
#include <stdio.h>
int main() {
    int n, i, j, a[100];
    scanf("%d", &n);
    for (i = 0; i < n; i++) scanf("%d", &a[i]);
    int element;
    scanf("%d", &element);
    int first = -1, last = -1;
    for (i = 0; i < n; i++) {
        if (a[i] == element) {
            if (first == -1) first = i;
            last = i;
        }
    }
    printf("First Occurrence: %d\n", first);
    printf("Last Occurrence: %d\n", last);
    return 0;
}
```

Input

5
1 2 3 4 2

Output

First Occurrence: 1
Last Occurrence: 4